

APPENDIX K

EXTENDED FUTURE OUTLOOK

A. Semantic Inter-Agent Protocols

While current communication is limited to state passing, future iterations will introduce intent-aware messaging. A shared queryable memory will support knowledge transfer with provenance tracking, reducing redundancy in itinerary generation.

B. Distributed Knowledge Sharing

We plan to enable learning components that leverage distributed observations. Local heuristics will be aggregated into global summaries, using structured push-pull mechanisms with conflict resolution to improve personalization without repeated refinements.

C. System Scalability

To address potential bottlenecks in the centralized supervisor, we propose a distributed design. This involves session affinity enforced by load balancers and a Redis-backed stateless execution model, allowing the system to scale across containerized deployments.